

Unit 2 Review

Real Numbers, Square Roots and Cube Roots **ANSWER KEY**

HOW TO USE THIS. This review covers every learning target from all six lessons of Unit 2 — more than the assessment does. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part.

PART A. Square Values and Square Roots

Lessons 1 and 2 · MA.8.AR.2.3 · 9 questions

1. List every perfect square from 1 to 225.

These are 1^2 through 15^2 . Write them out once and the rest of this unit gets easier.

Answer. 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225

2. Evaluate. $(-7)^2$

$(-7)(-7) = 49$. A negative times a negative is positive.

Answer. 49

3. Evaluate. $\sqrt{196}$

$14 \times 14 = 196$. The radical sign asks for the positive root only.

Answer. 14

4. Evaluate. $-\sqrt{169}$

$\sqrt{169} = 13$, and the minus sign in front makes it -13 . The minus is outside the radical.

Answer. -13

5. Evaluate. $\sqrt{-49}$

No real number times itself gives -49 . Positive $\times$ positive is positive and negative $\times$ negative is positive.

Answer. Not a real number.

6. A square patio covers 144 square feet. How long is each side?

Side $\times$ side = 144, so the side is $\sqrt{144} = 12$.

Answer. 12 feet

7. Explain the difference between $\sqrt{81}$ and the solutions of $x^2 = 81$.

The radical sign names one number, the principal square root, and it is never negative.

The equation asks which numbers square to 81, and there are two of them.

Answer. $\sqrt{81} = 9$. $x^2 = 81$ has two solutions, 9 and -9.

8. True or false: $\sqrt{225} = \pm 15$

Only $x^2 = 225$ has two answers. The radical sign by itself gives 15.

Answer. *False.* $\sqrt{225} = 15$.

9. **SSDD.** Every part below is about $\sqrt{40}$.

Every part is about the same number. They are not the same question — that is the whole item, and it is why the label is on this page and not on theirs.

- a. Between which two consecutive integers does $\sqrt{40}$ lie, and which is it closer to?

$36 < 40 < 49$, so it lies between $\sqrt{36} = 6$ and $\sqrt{49} = 7$.

$40 - 36 = 4$ and $49 - 40 = 9$, so 40 sits nearer 36.

Answer. *Between 6 and 7, and closer to 6.*

- b. Is $\sqrt{40}$ rational or irrational? How do you know?

40 is not a perfect square, so its square root is a decimal that never stops and never repeats.

Answer. *Irrational.*

- c. Solve $x^2 = 40$. How many solutions does it have?

The equation asks for every number whose square is 40, and both $\sqrt{40}$ and $-\sqrt{40}$ qualify.

This is the one part with two answers. Part a has one number and part b has one word.

Answer. $x = \pm\sqrt{40}$. *Two solutions.*

- d. $\sqrt{40} \approx 6.3$. Explain why $\sqrt{40} \times \sqrt{40}$ is exactly 40 but 6.3×6.3 is not.

$6.3 \times 6.3 = 39.69$, which is not 40.

$\sqrt{40}$ is the number that squares to 40 exactly. 6.3 is only the nearest tenth to it.

Answer. *Because 6.3 is rounded and the root is exact.*

Part a estimates, part b classifies, part c solves and counts solutions, and part d is the exact-versus-rounded distinction. Only part c has two answers, and only part d needs no value for the root at all.

PART B. Cube Values and Cube Roots

Lesson 3 · MA.8.AR.2.3 · 6 questions

10. List every perfect cube from -125 to 125 .

These are $(-5)^3$ through 5^3 . Unlike squares, the list runs through the negatives too.

Answer. $-125, -64, -27, -8, -1, 0, 1, 8, 27, 64, 125$

11. Evaluate. $(-4)^3$

$(-4)(-4) = 16$, then $16(-4) = -64$. Three negative factors leave a negative.

Answer. -64

12. Evaluate. $\sqrt[3]{125}$

$5 \times 5 \times 5 = 125$.

Answer. 5

13. Evaluate. $\sqrt[3]{-27}$

$(-3)(-3)(-3) = -27$.

Answer. -3

14. A cube holds 64 cubic centimeters. How long is each edge?

Edge $\times$ edge $\times$ edge = 64, so the edge is $\sqrt[3]{64} = 4$.

Answer. 4 centimeters

15. Why does $\sqrt[3]{-64}$ have a real value while $\sqrt{-64}$ does not?

$\sqrt[3]{-64} = -4$ because $(-4)(-4)(-4) = -64$.

Nothing squares to a negative: positive $\times$ positive and negative $\times$ negative are both positive.

Answer. *A number multiplied by itself three times can be negative, but a number multiplied by itself twice cannot.*

PART C. Solving $x^2 = p$ and $x^3 = p$

Lessons 2 and 3 · MA.8.AR.2.3 · 6 questions

16. Solve. $x^2 = 121$

$\sqrt{121} = 11$, and -11 squares to 121 as well. Both are solutions.

Answer. $x = 11$ and $x = -11$

17. Solve. $x^2 = 0$

$0 \times 0 = 0$, and no other number squares to 0. This is the one square equation with a single solution.

Answer. $x = 0$ only

18. Solve. $x^2 = -16$

Squaring any real number gives a positive value or zero, never -16 .

Answer. No real solutions.

19. Solve. $x^3 = 27$

$3 \times 3 \times 3 = 27$. Cube equations have exactly one real solution.

Answer. $x = 3$

20. Solve. $x^3 = -125$

$(-5)(-5)(-5) = -125$.

Answer. $x = -5$

21. Select all of the solutions of $x^2 = 100$.

☐ $x = -100$

☒ $x = -10$

☒ $x = 10$

☐ $x = 100$

-100 and 100 square to $10\,000$, not 100 . This is the most common wrong pair on this item type.

PART D. Approximating Irrational Numbers

Lesson 4 · MA.8.NSO.1.1 · 6 questions

22. Between which two whole numbers does $\sqrt{30}$ lie? Which is it closer to?

$25 < 30 < 36$, so $\sqrt{30}$ is between 5 and 6.

$30 - 25 = 5$ and $36 - 30 = 6$, so 30 sits nearer 25.

Answer. Between 5 and 6, closer to 5.

23. Between which two whole numbers does $\sqrt[3]{70}$ lie? Which is it closer to?

$64 < 70 < 125$, so $\sqrt[3]{70}$ is between 4 and 5.

$70 - 64 = 6$ and $125 - 70 = 55$, so 70 sits much nearer 64.

Answer. Between 4 and 5, closer to 4.

24. Approximate $\sqrt{52}$ to the nearest tenth.
 $49 < 52 < 64$, so the answer starts with 7.
 $7.2 \times 7.2 = 51.84$ and $7.3 \times 7.3 = 53.29$. 51.84 is closer to 52.
Answer. 7.2
25. Approximate $\sqrt[3]{20}$ to the nearest tenth.
 $8 < 20 < 27$, so the answer starts with 2.
 $2.7^3 = 19.683$ and $2.8^3 = 21.952$. 19.683 is closer to 20.
Answer. 2.7
26. Approximate 2π to the nearest tenth.
 $\pi \approx 3.14$, so $2\pi \approx 6.28$, which rounds to 6.3.
Answer. 6.3
27. Dev uses $\sqrt{11} \approx 3.3$ to approximate $4\sqrt{11}$ and gets 13.2. A calculator gives 13.3. What went wrong?
 $\sqrt{11} = 3.3166\dots$, and $4 \times 3.3166 = 13.266$, which rounds to 13.3.
Rounding to 3.3 first threw away 0.0166, and multiplying by 4 turned that into 0.066 — enough to change the tenths digit.
Answer. He rounded first. Round last.

PART E. Rational or Irrational

Lessons 4 and 5 · MA.8.NSO.1.1 · 6 questions

28. Is $\sqrt{49}$ rational or irrational? Explain.
49 is a perfect square, so its root is a whole number, and every whole number is rational.
Answer. Rational. It equals 7.
29. Is $\sqrt[3]{10}$ rational or irrational? Explain.
10 is not a perfect cube — it sits between 8 and 27 — so the root never terminates and never repeats.
Answer. Irrational.

30. Is $0.\overline{4}$ rational or irrational? Explain.

A repeating decimal can always be written as a fraction, and a fraction of whole numbers is rational.

Answer. *Rational. It equals $\frac{4}{9}$.*

31. Is -0.5π rational or irrational? Explain.

π is irrational, and multiplying an irrational number by a non-zero rational number leaves it irrational.

Answer. *Irrational.*

32. Is $\frac{\sqrt{144}}{4}$ rational or irrational? Explain.

$\sqrt{144} = 12$ and $12 \div 4 = 3$. A radical in an expression does not make the expression irrational.

Answer. *Rational. It equals 3.*

33. Sort the six numbers into the table.

$$\sqrt{36}, \sqrt{37}, \frac{22}{7}, \pi, \sqrt[3]{8}, -\sqrt[3]{9}$$

Rational	Irrational
$\sqrt{36}$ $\frac{22}{7}$ $\sqrt[3]{8}$	$\sqrt{37}$ π $-\sqrt[3]{9}$

22/7 is the trap. It is a common approximation of π , but it is a fraction of whole numbers, so it is rational. π itself is not.

PART F. Plotting, Ordering and Comparing

Lessons 5 and 6 · MA.8.NSO.1.2 · 8 questions

34. Order from least to greatest. $\sqrt{17}, -\sqrt[3]{30}, 4.6, -3.5, \frac{13}{4}$

$-3.5, -\sqrt[3]{30} \approx -3.1, 13/4 = 3.25, \sqrt{17} \approx 4.1, 4.6$.

$27 < 30 < 64$ puts $\sqrt[3]{30}$ between 3 and 4, near 3.1. $16 < 17 < 25$ puts $\sqrt{17}$ just past 4.

Answer. *$-3.5, -\sqrt[3]{30}, \frac{13}{4}, \sqrt{17}, 4.6$*

35. Order from least to greatest. $-\sqrt{50}$, -2π , $\sqrt[3]{-20}$, $\frac{19}{4}$, $\sqrt{35}$

$$-\sqrt{50} \approx -7.1, -2\pi \approx -6.3, \sqrt[3]{-20} \approx -2.7, 19/4 = 4.75, \sqrt{35} \approx 5.9.$$

With negatives, the bigger the root the further left it sits.

Answer. $-\sqrt{50}$, -2π , $\sqrt[3]{-20}$, $\frac{19}{4}$, $\sqrt{35}$

36. Compare using $<$, $>$ or $=$. $\sqrt{26}$ 5.2

$$5.2 \times 5.2 = 27.04, \text{ and } 26 < 27.04, \text{ so } \sqrt{26} < 5.2. \text{ No estimating needed.}$$

Answer. $\sqrt{26} < 5.2$

37. Compare using $<$, $>$ or $=$. $-\sqrt[3]{65}$ -4

$$65 > 64, \text{ so } \sqrt[3]{65} > 4. \text{ Putting a minus sign on both flips the comparison: } -\sqrt[3]{65} < -4.$$

Answer. $-\sqrt[3]{65} < -4$

38. Compare using $<$, $>$ or $=$. 4π $\sqrt{150}$

$$4\pi \approx 12.6. 144 < 150 < 169, \text{ so } \sqrt{150} \text{ is just past 12, about 12.2.}$$

Answer. $4\pi > \sqrt{150}$

39. Compare using $<$, $>$ or $=$. $\sqrt[3]{-8} \cdot 5$ $-\sqrt{100}$

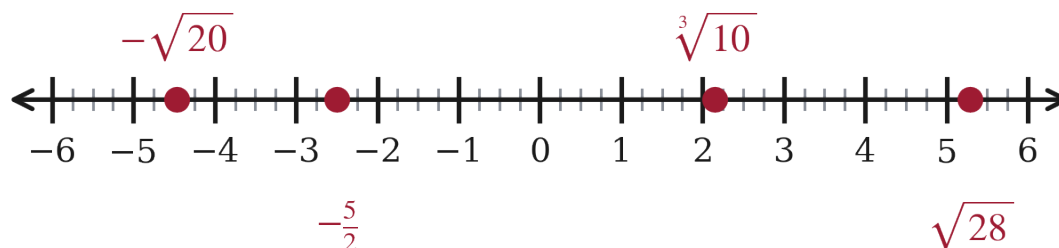
$$\sqrt[3]{-8} = -2 \text{ and } (-2)(5) = -10. -\sqrt{100} = -10. \text{ They are the same number.}$$

Answer. $\sqrt[3]{-8} \cdot 5 = -\sqrt{100}$

40. Plot and label each value. $-\sqrt{20}$, $-\frac{5}{2}$, $\sqrt[3]{10}$, $\sqrt{28}$

$$-\sqrt{20} \approx -4.5, -5/2 = -2.5, \sqrt[3]{10} \approx 2.2, \sqrt{28} \approx 5.3.$$

Answer. *See the number line.*



41. Marcus orders $-\sqrt[3]{30}$, -3.2 , $-\sqrt{8}$ from least to greatest and writes $-\sqrt{8}$, -3.2 , $-\sqrt[3]{30}$. Is he right? If not, fix it.

$-\sqrt[3]{30} \approx -3.1$, -3.2 , $-\sqrt{8} \approx -2.8$.

Marcus wrote the given list BACKWARDS instead of ordering it — reversing a list only helps if it was already in the opposite order, and this one was not. (Ordering by size and dropping the signs would have given $-\sqrt{8}$, $-\sqrt[3]{30}$, -3.2 , which is not what he wrote either.) With negatives the number that looks biggest is the smallest: -3.2 has the largest size and sits furthest left.

Answer. *No.* -3.2 , $-\sqrt[3]{30}$, $-\sqrt{8}$